

Lab05 Report

Sample: smoke_test
Frames: 6

--- Dashboard ---
Detected cells: 14
Tracks: 4
Divisions: 1
Avg. lifetime: 3.5 frames
Avg. velocity: 3.747 px/frame

Method

Cells were detected per frame via Gaussian smoothing ($\sigma=2$) followed by local maximum detection (threshold=800, min. distance=12px) on each frame's maximum-intensity projection. Candidate detections were classified by a trained CNN (CellCNN). Frame-to-frame linking used optimal (Hungarian) bipartite assignment (max. distance=25px); cell divisions were detected via a second assignment pass on unmatched detections (max. distance=37.5px), each daughter assigned a new track ID linked to its parent.

Tracking Metrics

Tracks: 4

Divisions: 1

Avg. lifetime: 3.5 frames

Avg. velocity: 3.747 px/frame

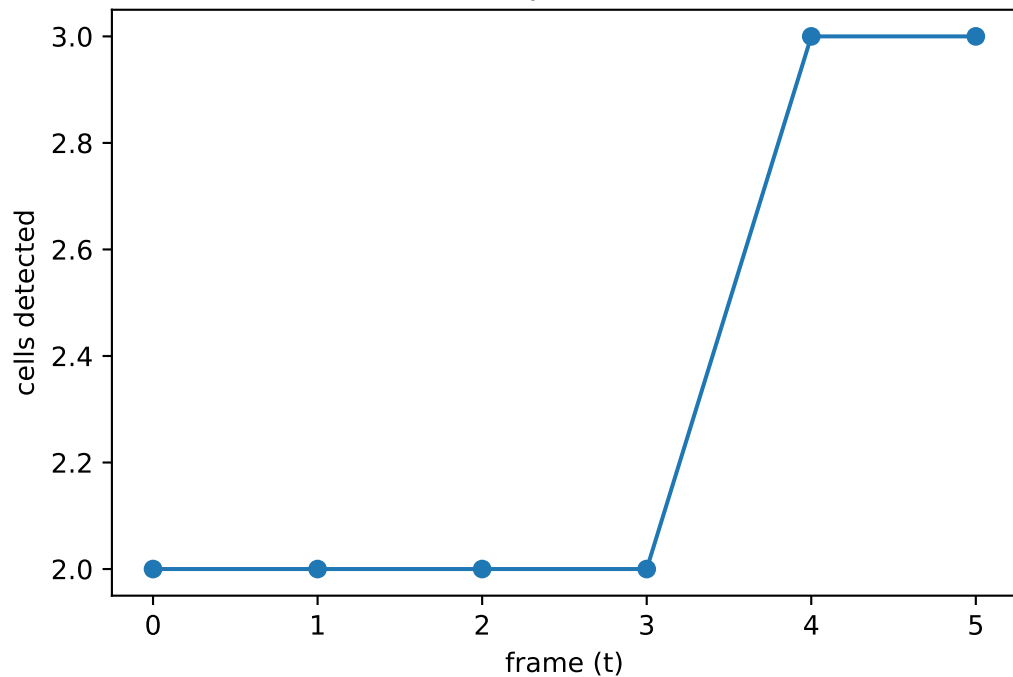
No ground truth / no benchmark comparison provided for this run – pass `benchmark_rows` (from `src/benchmark.py`) for a Method vs. Method comparison with precision/recall/runtime/memory.

Division Analysis

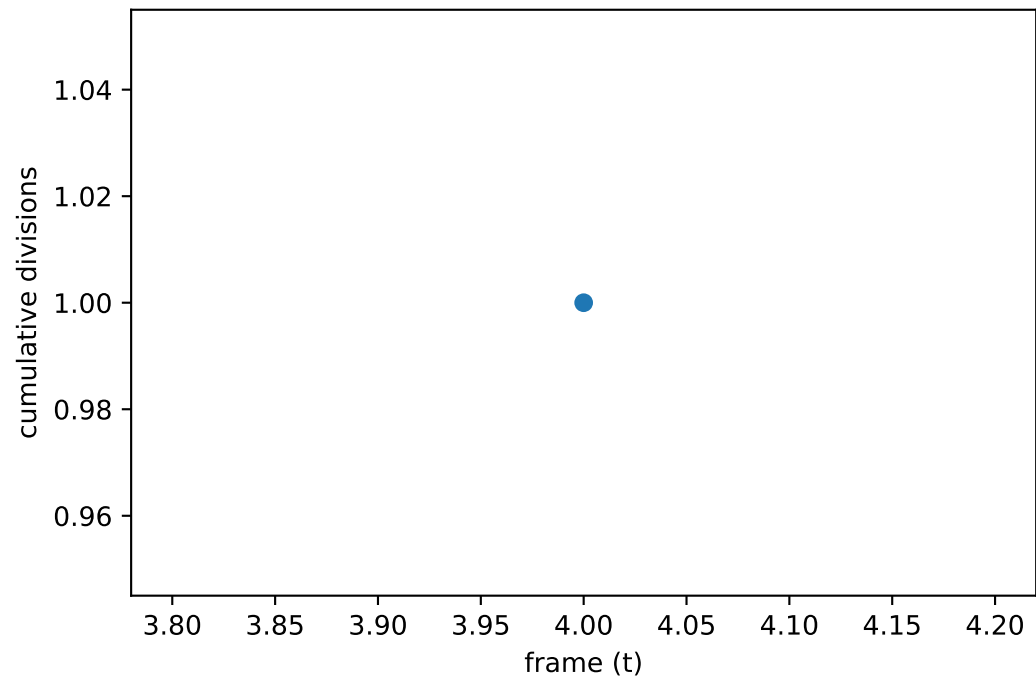
1 division(s) detected:

1. Frame 4: track 1 (alive frames 0-3) -> tracks 3, 4
 track 3: alive frames 4-5 (2 frames)
 track 4: alive frames 4-5 (2 frames)

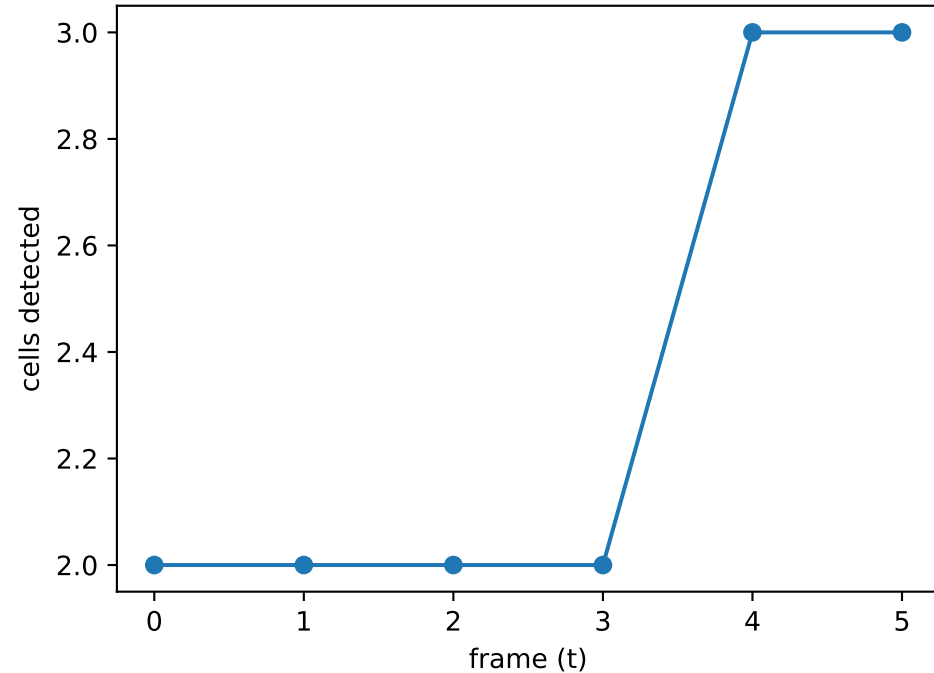
Cells per frame



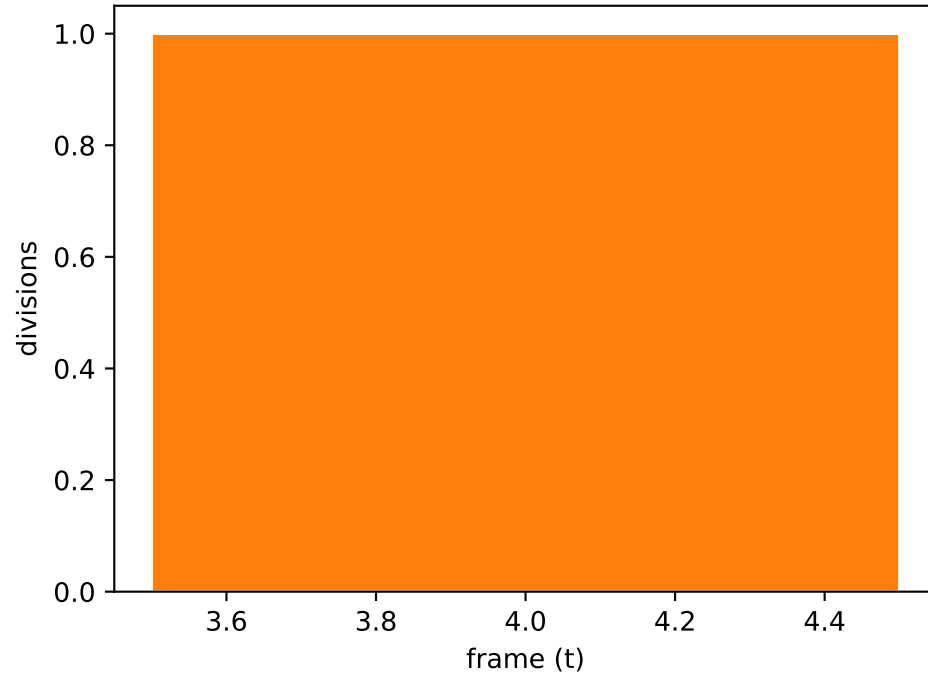
Divisions over time (total: 1)



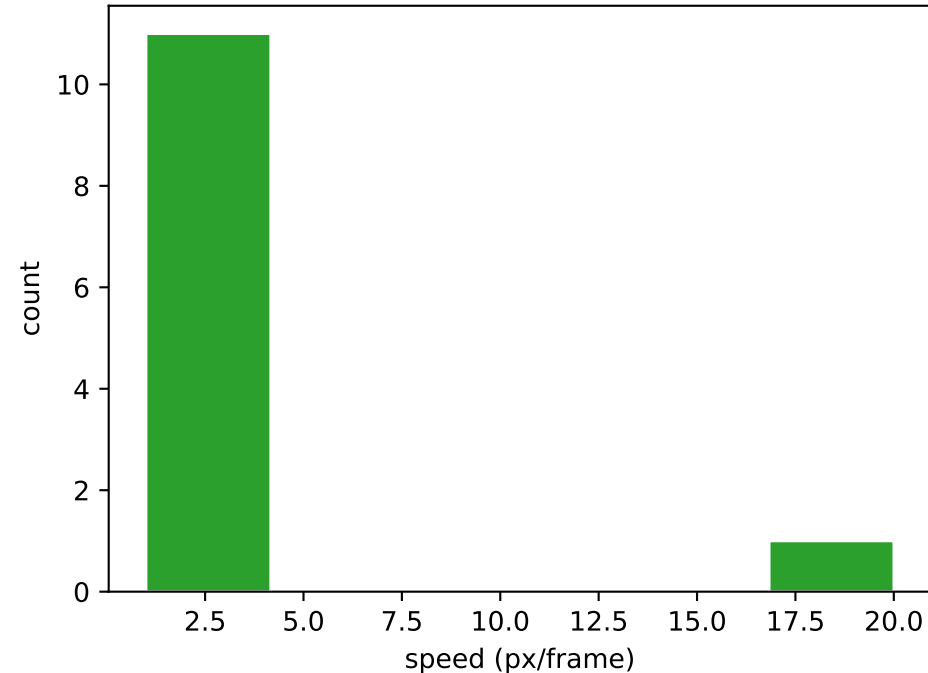
Cell count over time



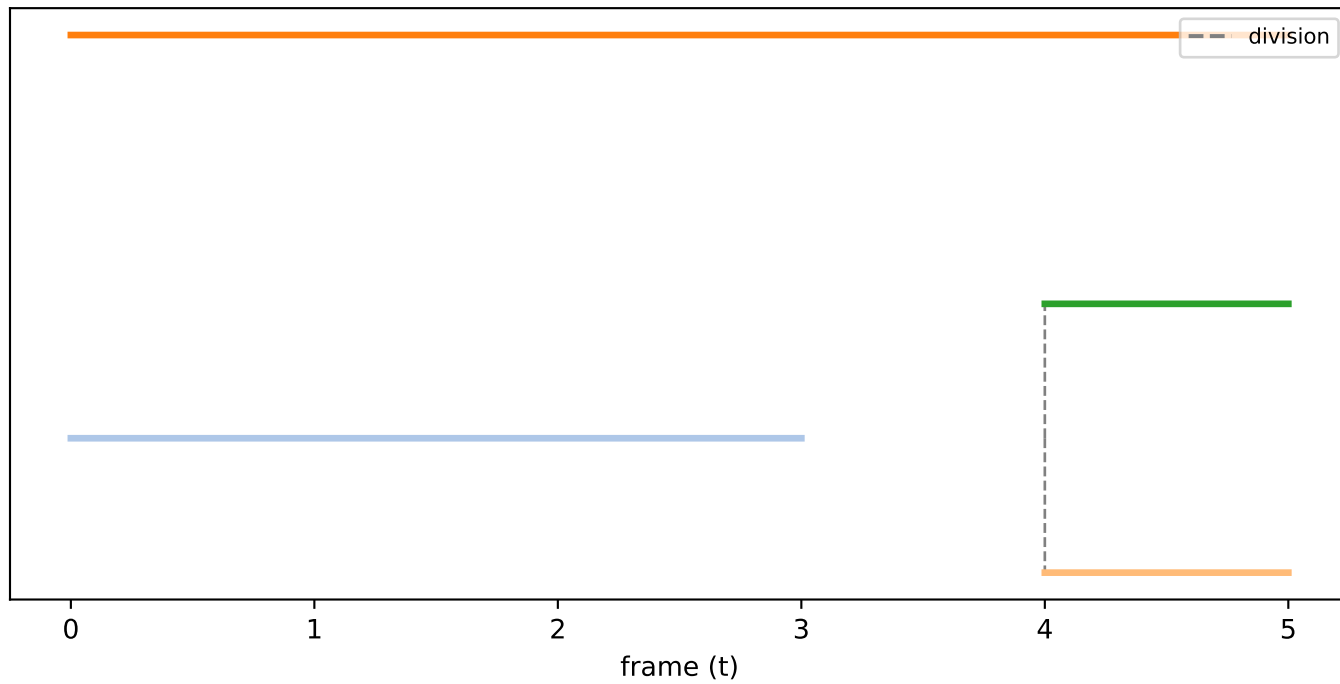
Divisions over time (total: 1)



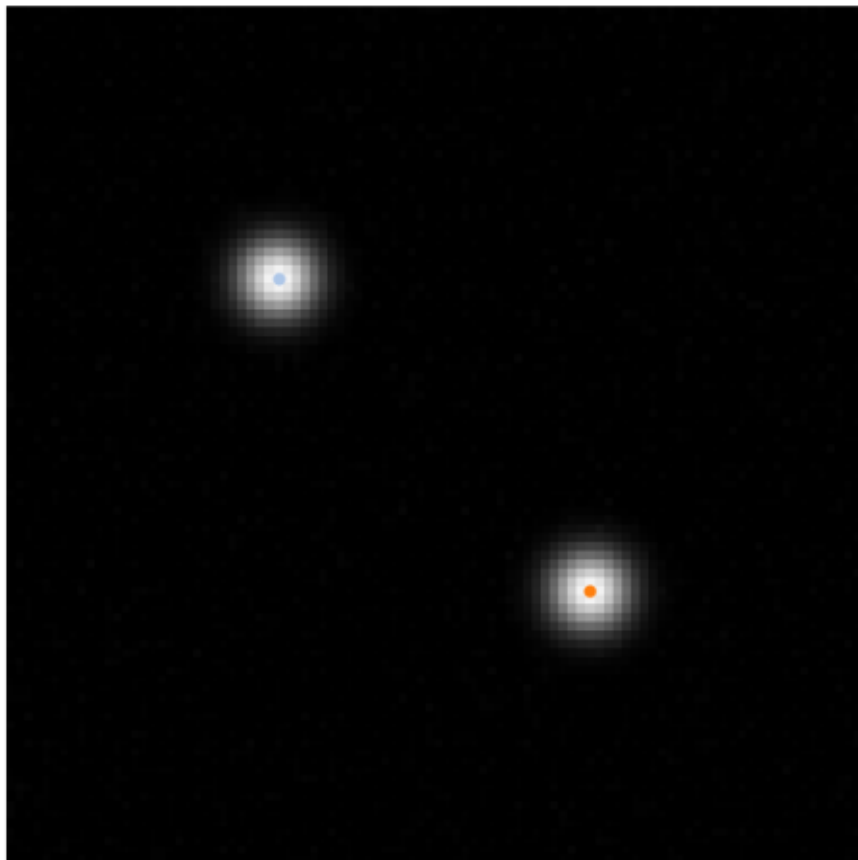
Velocity distribution



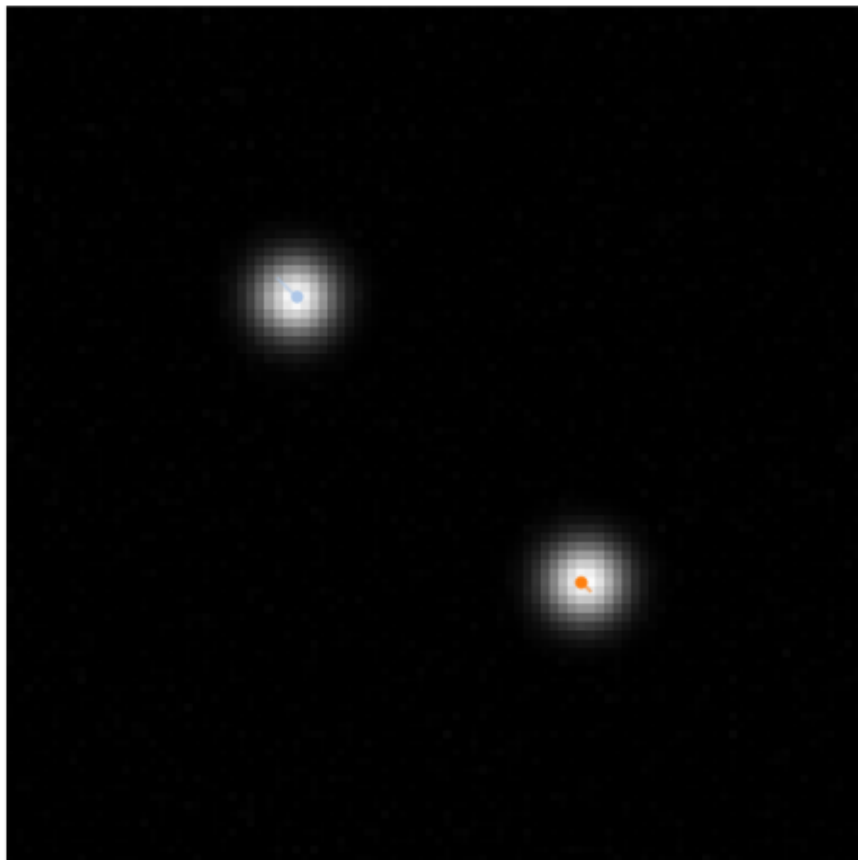
Lineage tree — 4 tracks, 1 division(s)



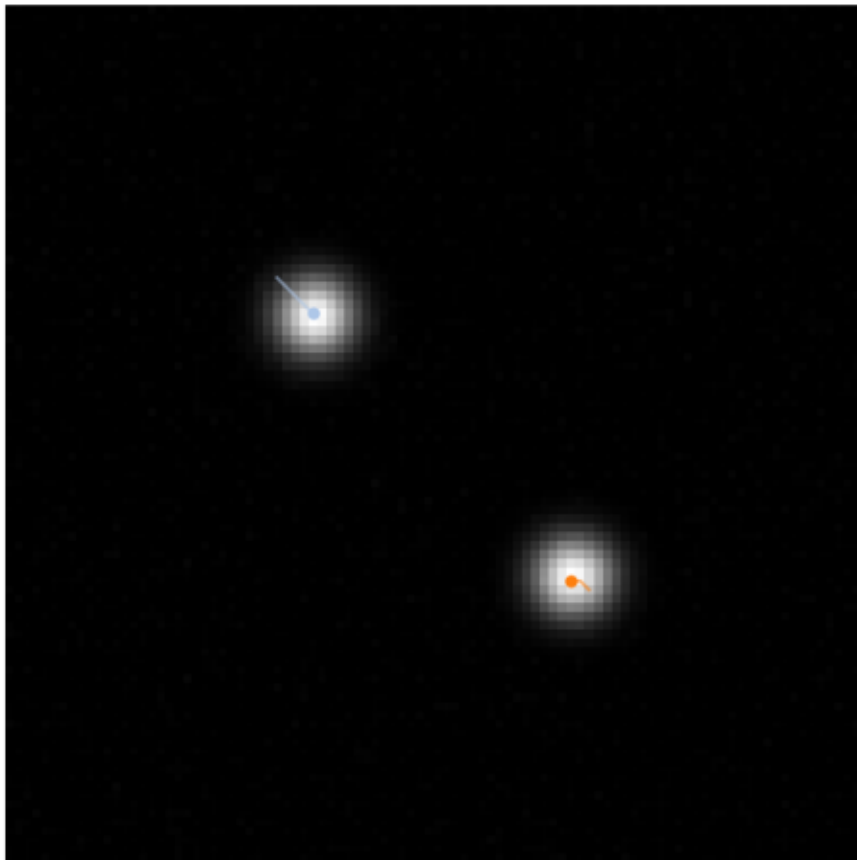
$t = 0$ (2 tracks visible, 0 division(s) this frame)



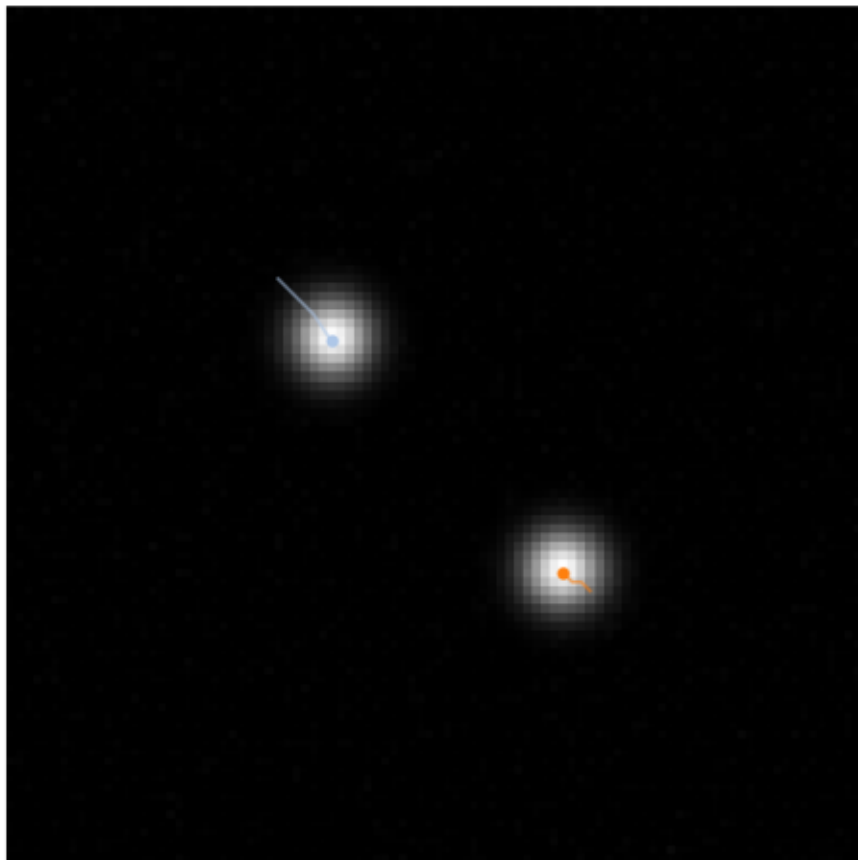
t = 1 (2 tracks visible, 0 division(s) this frame)



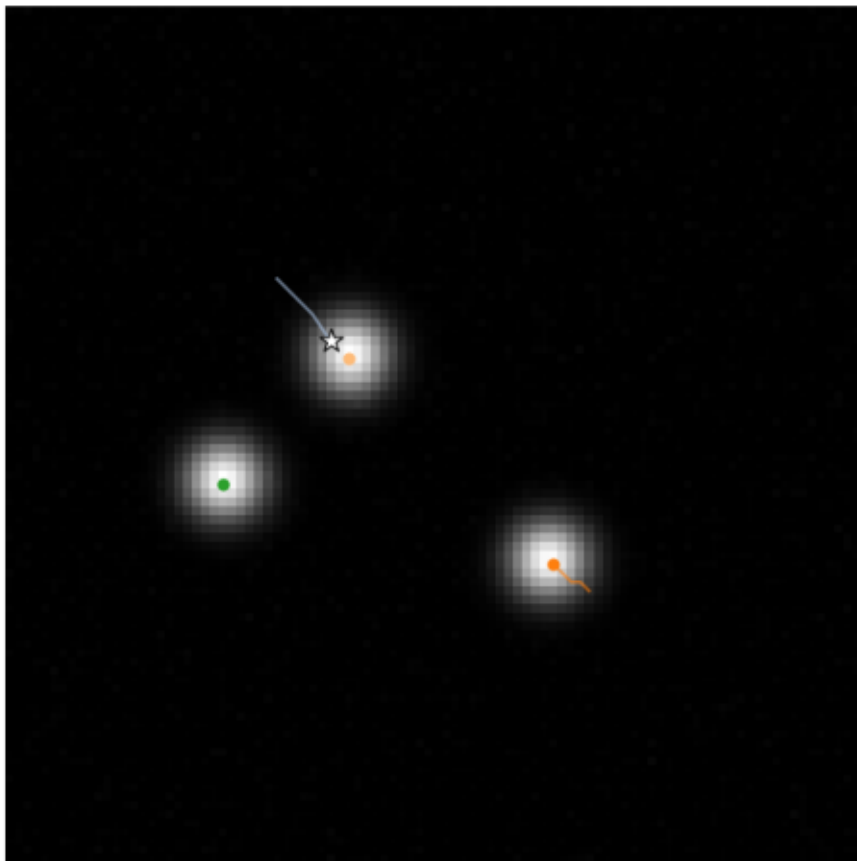
$t = 2$ (2 tracks visible, 0 division(s) this frame)



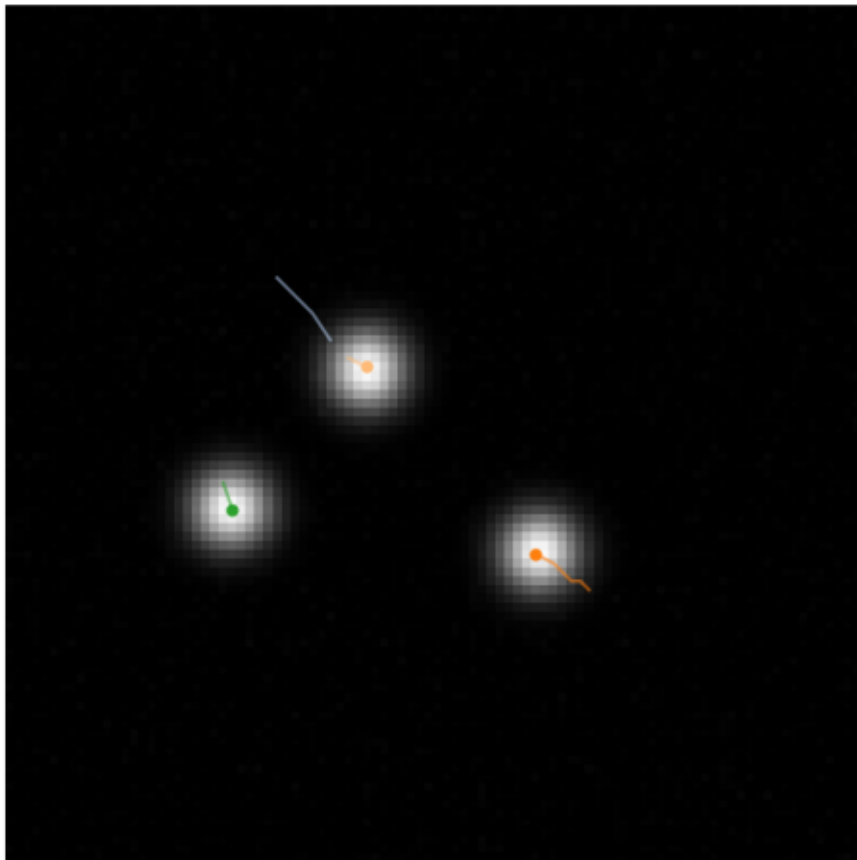
t = 3 (2 tracks visible, 0 division(s) this frame)



$t = 4$ (4 tracks visible, 1 division(s) this frame)



$t = 5$ (4 tracks visible, 0 division(s) this frame)



Limitations

- Detection uses a fixed threshold (`config.DETECTION_THRESHOLD`) not recalibrated per-dataset; brightness differences between microscopes/samples can silently change detection sensitivity (see `mvp/make_test_data.py`'s comments for a concrete example of this happening in practice).
- Detection and tracking both operate on the maximum-intensity projection of each 3D frame, not the full 3D volume – z-position information is not tracked.
- Division detection (`src/lineage.py`) uses a distance-based heuristic (a second assignment pass on unmatched detections), not a learned model – it can miss divisions where daughters land far apart, or misattribute a division in dense regions where multiple cells are within range of the same parent (see `src/lineage.py`'s docstring).
- No motion prediction (e.g. a Kalman filter) – occluded or temporarily undetected cells will appear as a track ending and a new one starting, not a continuous track with a gap.
- Speed/velocity are reported in pixels/frame unless a real physical calibration (pixel size + frame interval) is set for this dataset's actual acquisition parameters – see this report's Tracking Metrics section for which applies here.
- THIS RUN: physical calibration was NOT set – all speed/lifetime numbers in this report are in pixels/frame, not real-world units.

Appendix: Run Configuration

GAUSSIAN_SIGMA = 2
DETECTION_THRESHOLD = 800
CELL_RADIUS = 12
TRACKING_MAX_DISTANCE = 25
DIVISION_MAX_DISTANCE = 37.5
PIXEL_SIZE_UM = None
FRAME_INTERVAL_MIN = None

Appendix: Track Summary

track_id	parent_track_id	children_track_ids	birth_frame	death_frame	track_length	avg_speed
1	NaN	[3, 4]	0	3	4	3.087469
2	NaN	[]	0	5	6	1.495742
3	1.0	[]	4	5	2	2.532248
4	1.0	[]	4	5	2	11.581139